



Engineering Electromagnetics

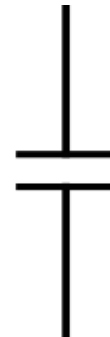
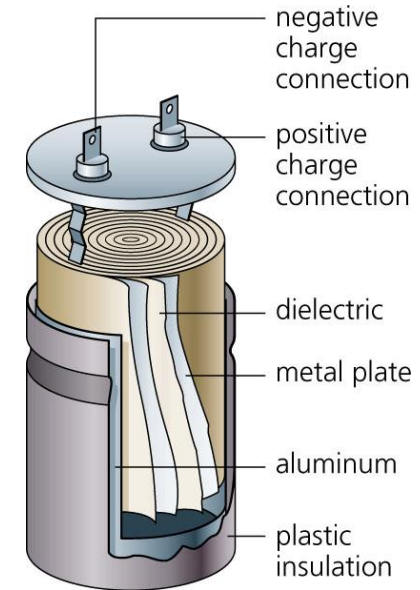
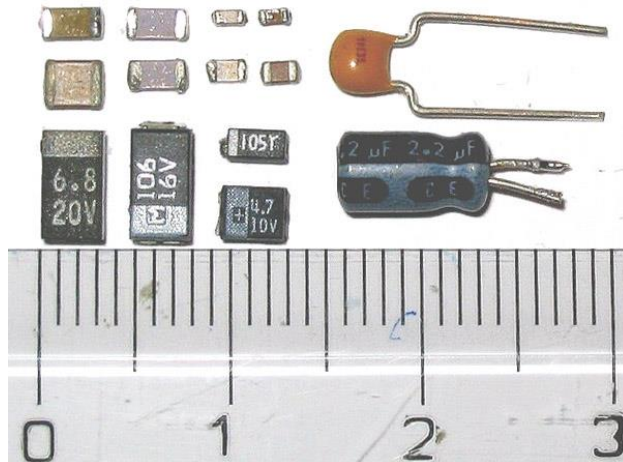
工程电磁场

Youwei Jia 嘉有为

10: Capacitance



- Capacitor Vs Capacitance 电容 (器) ? 电容?



Fixed Capacitor



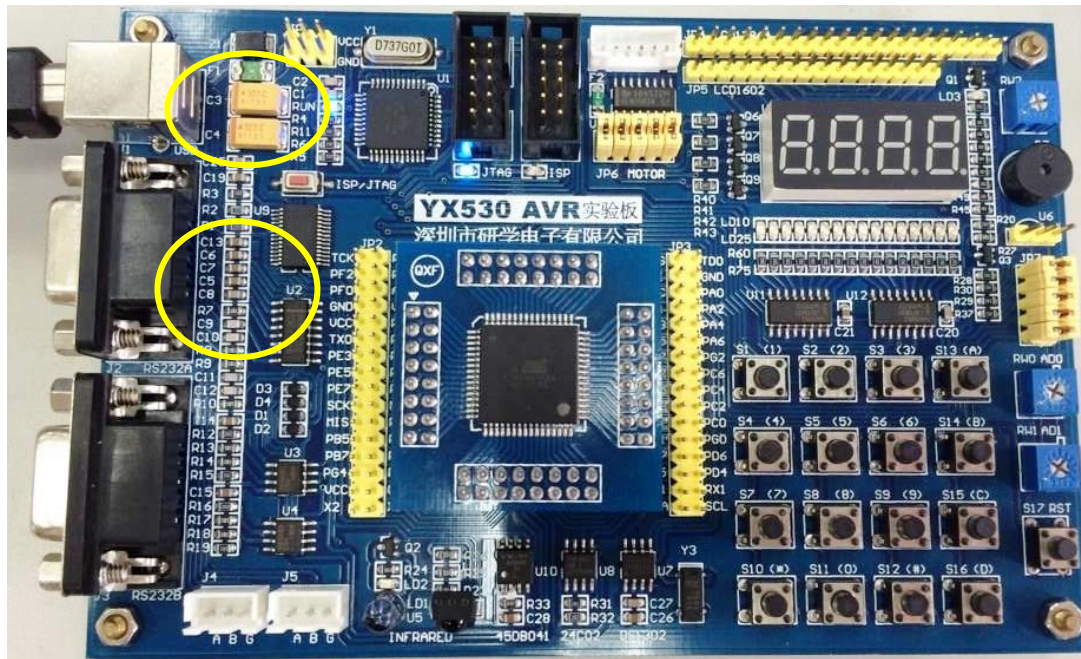
Polarized Capacitor



Variable Capacitor

10: Capacitance

- Capacitor Vs Capacitance 电容 (器) ? 电容?



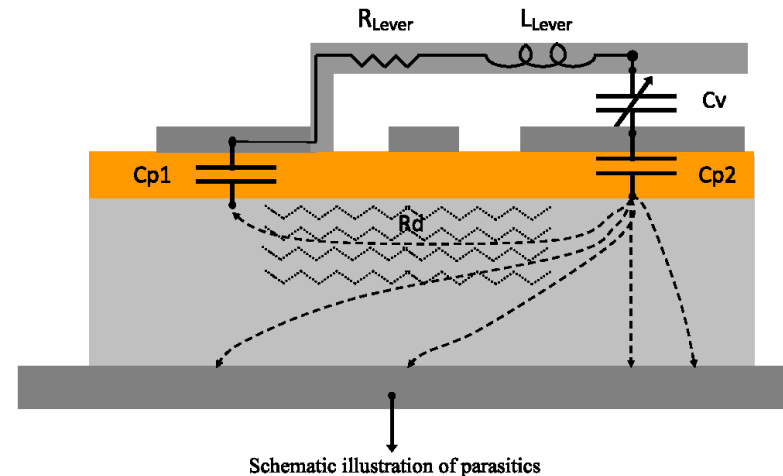
A capacitor is a electrical component / device 元器件 used to store electric energy (electric flux, electric field, electric charge).

Most of time, capacitors are purposely designed to perform specific functions, but sometimes, capacitance may exist as an unavoidable by-product of the device structure. Parasitical Capacitance 寄生电容

10: Capacitance

- Capacitor Vs Capacitance 电容 (器) ? 电容?

Parasitical capacitance in semiconductor devices



Capacitance is a **physical quantity** used to measure the capability of energy storage in electrical devices (Capacitors).

How to define capacitance, and why?

How to calculate the capacitance of a electrical device?

How to design a capacitor to fulfill specific purpose?

How to eliminate the bad effects aroused by parasitical capacitance?

What are the key factors that can affect the capacitance of capacitors?

10: Capacitance

- Capacitance defined

A simple capacitor consists of **two oppositely charged conductors** surrounded by a **uniform dielectric**.

An increase in Q by some factor results in an increase in $\mathbf{E}$ (and in $\mathbf{D}$) by the same factor.

where $Q = \oint_S \mathbf{D} \cdot d\mathbf{S}$

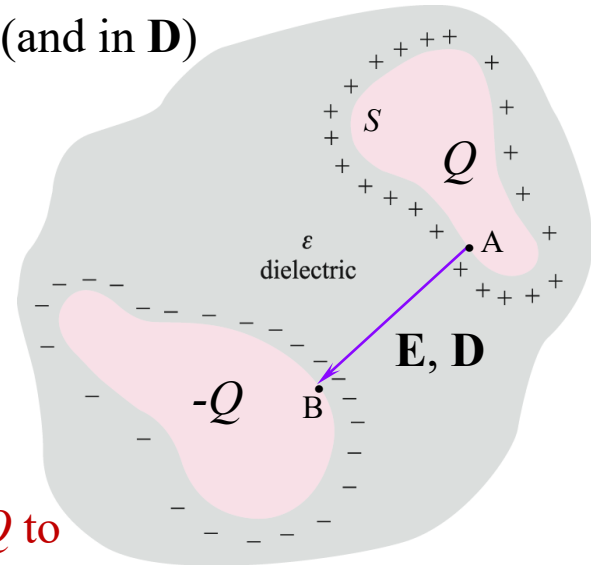
Consequently, the potential difference between conductors:

$$V_0 = - \int_B^A \mathbf{E} \cdot d\mathbf{L}$$

will also increase by the same factor -- so the ratio of Q to V_0 is a constant. We define the *capacitance* of the structure as the ratio of the stored charge to the applied voltage, or

$$C = \frac{Q}{V_0}$$

Units are **Coul/V** or **Farads**



10: Capacitance

- Capacitance defined

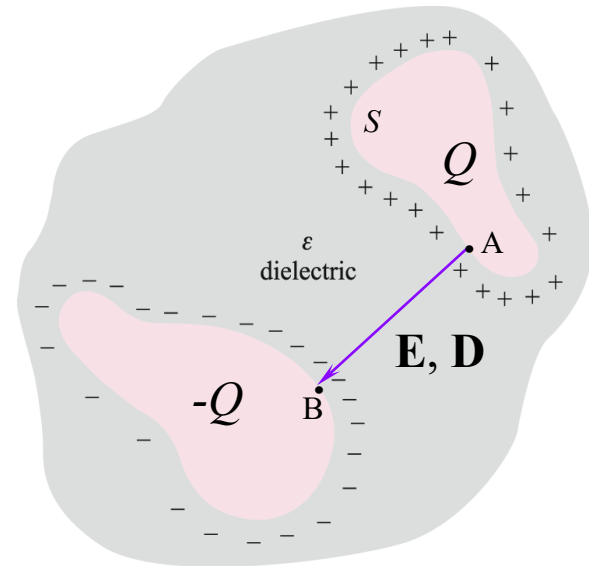
In general terms, we determine Q by a surface **integral over the positive conductors**, and we find V_0 by **carrying a unit positive charge from the negative to the positive surface**,

$$C = \frac{\oint_S \epsilon \mathbf{E} \cdot d\mathbf{S}}{-\int_{-}^{+} \mathbf{E} \cdot d\mathbf{L}}$$

The capacitance **is independent of the potential and total charge**, for their ratio is constant.

The capacitance is **a function only of the physical dimensions of the system of conductors and of the permittivity of the homogeneous dielectric**.

Capacitance is measured in *farads* (F), where a farad is defined as one coulomb per volt. Common values of capacitance are apt to be very small fractions of a farad, and consequently more practical units are the **microfarad (μF)**, the **nanofarad (nF)**, and the **picofarad (pF)**.

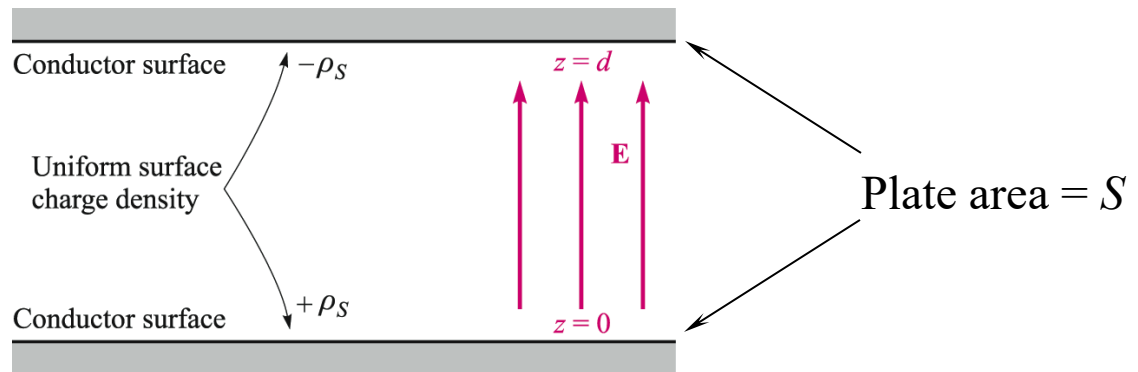


10: Capacitance



- Parallel-Plane Capacitor

The horizontal dimensions are assumed to be much greater than the plate separation, d . Therefore, electric field can be assumed to lie only in the z direction, and potential varies only with z .



Apply the boundary for $\mathbf{D}$ at the surface of a perfect conductor:

$$\text{Lower plate: } \mathbf{D} \cdot \mathbf{n}_\ell \Big|_{z=0} = \mathbf{D} \cdot \mathbf{a}_z = \rho_s \Rightarrow \underline{\mathbf{D} = \rho_s \mathbf{a}_z}$$

$$\text{Upper Plate: } \mathbf{D} \cdot \mathbf{n}_u \Big|_{z=d} = \mathbf{D} \cdot (-\mathbf{a}_z) = -\rho_s \Rightarrow \underline{\mathbf{D} = \rho_s \mathbf{a}_z}$$

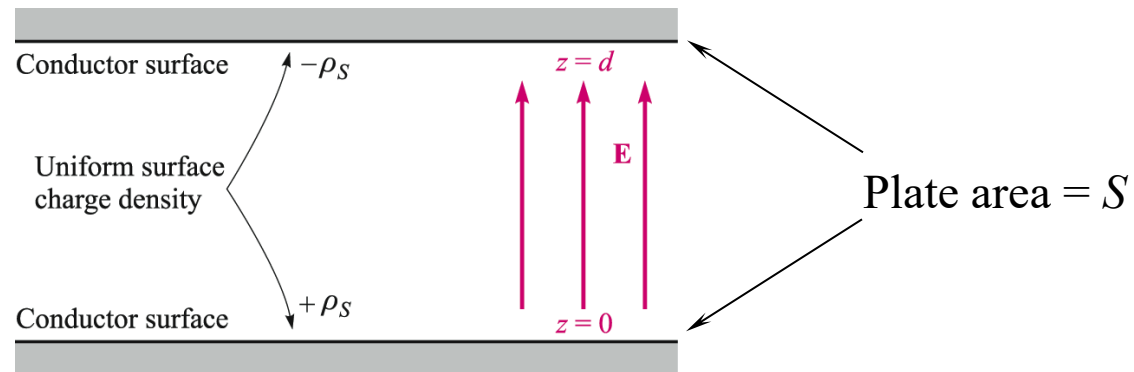
} Same result either way!

The electric field between plates is therefore:

$$\mathbf{E} = \frac{\rho_s}{\epsilon} \mathbf{a}_z$$

10: Capacitance

- Capacitance of Parallel-Plane Capacitor



Now with $\mathbf{E} = \frac{\rho_S}{\epsilon} \mathbf{a}_z$ The voltage between plate can be found through:

$$V_0 = - \int_{\text{upper}}^{\text{lower}} \mathbf{E} \cdot d\mathbf{L} = - \int_d^0 \frac{\rho_S}{\epsilon} dz = \frac{\rho_S}{\epsilon} d$$

Then with $Q = \rho_S S$ we finally obtain

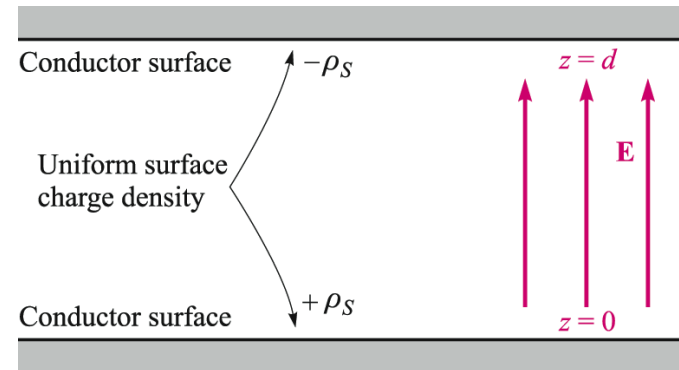
$$C = \frac{Q}{V_0} = \frac{\epsilon S}{d}$$

10: Capacitance

- Energy Stored in Parallel-Plane Capacitor

$$W_E = \int_{vol} \frac{1}{2} \mathbf{D} \cdot \mathbf{E} dv$$

or



$$W_E = \frac{1}{2} \int_{vol} \epsilon E^2 dv = \frac{1}{2} \int_S \int_0^d \frac{\epsilon \rho_S^2}{\epsilon^2} dz dS = \frac{1}{2} \frac{\rho_S^2}{\epsilon} S d = \frac{1}{2} \underbrace{\frac{\epsilon S}{d}}_C \underbrace{\frac{\rho_S^2 d^2}{\epsilon^2}}_{V_0^2}$$

There are three ways of writing the energy:

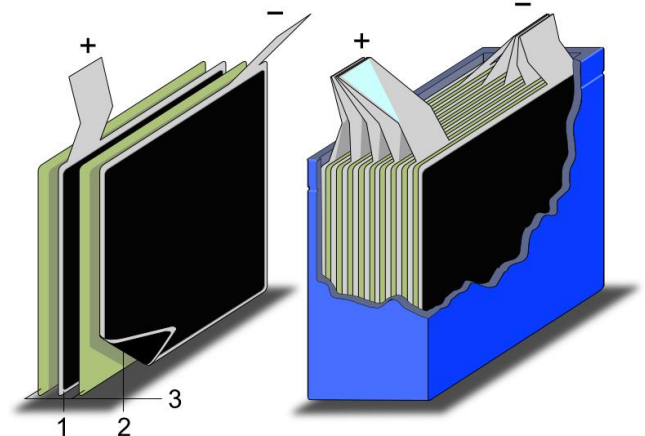
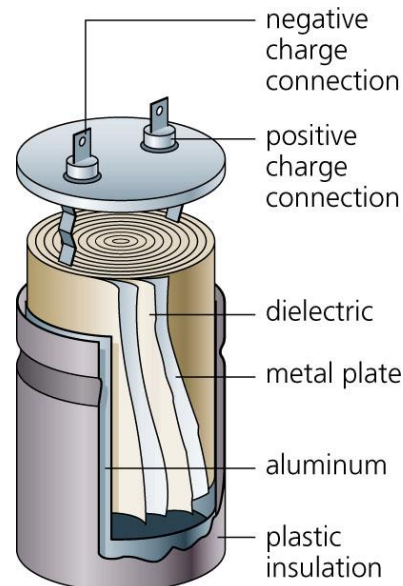
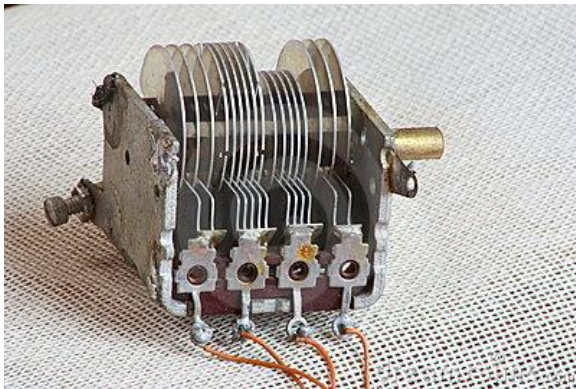
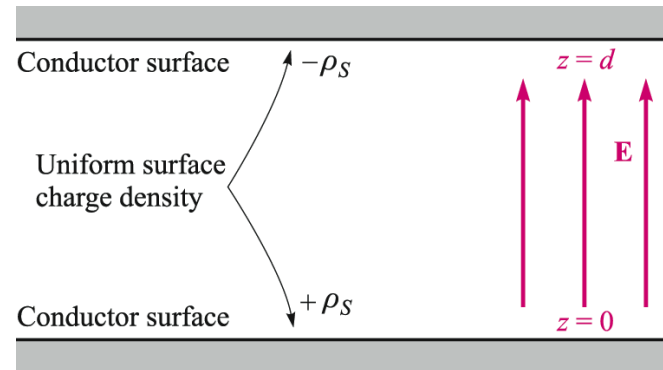
$$W_E = \frac{1}{2} C V_0^2 = \frac{1}{2} Q V_0 = \frac{1}{2} \frac{Q^2}{C}$$

10: Capacitance



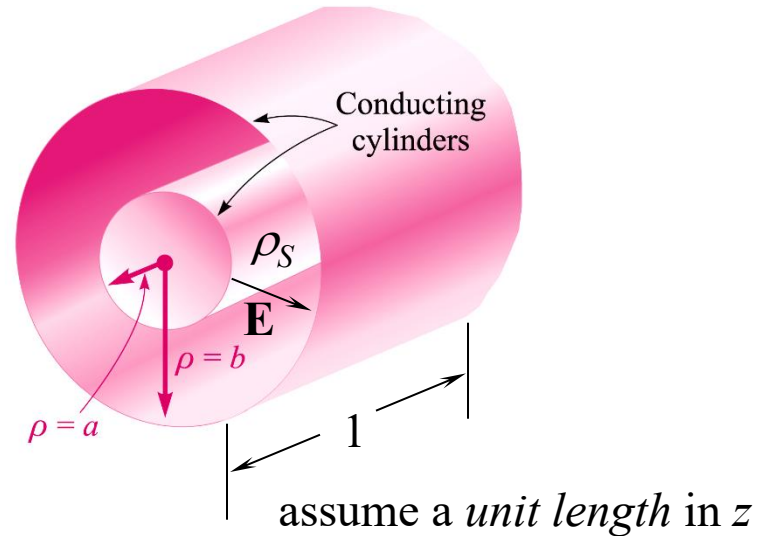
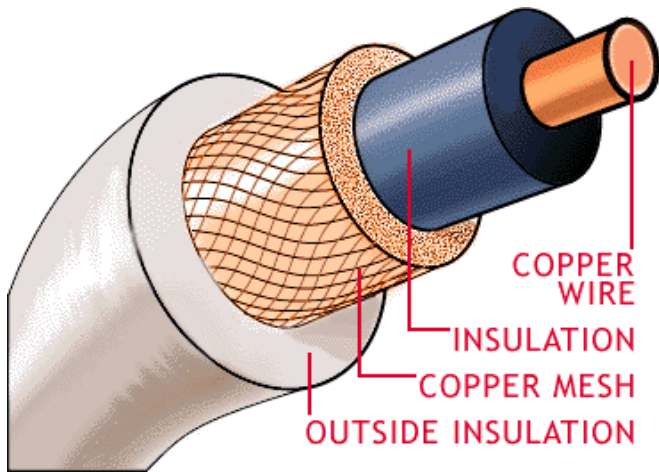
- Parallel-Plane Capacitors in practical application

$$C = \frac{Q}{V_0} = \frac{\epsilon S}{d}$$



10: Capacitance

- Capacitance in Coaxial Transmission Line



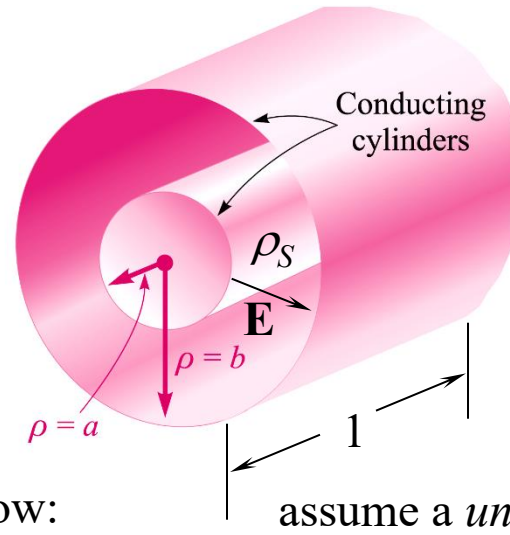
10: Capacitance

• Capacitance in Coaxial Transmission Line

We previously found using Gauss' Law:

$$\mathbf{E}(\rho) = \frac{a\rho_s}{\epsilon\rho} \mathbf{a}_\rho \text{ V/m} \quad (a < \rho < b)$$

$\mathbf{E} = 0$ elsewhere, assuming a hollow inner conductor, and equal and opposite charges on the inner and outer conductors.



The potential difference between conductors is now:

$$V_0 = - \int_b^a \mathbf{E} \cdot d\mathbf{L} = - \int_b^a \frac{a\rho_s}{\epsilon\rho} \mathbf{a}_\rho \cdot \mathbf{a}_\rho d\rho = \frac{a\rho_s}{\epsilon} \ln \left(\frac{b}{a} \right)$$

...and the charge on the inner conductor per unit length is:

$$Q = \underline{2\pi a(1)\rho_s}$$

Finally:

$$C = \frac{Q}{V_0} = \frac{2\pi\epsilon}{\ln(b/a)} \text{ F/m}$$

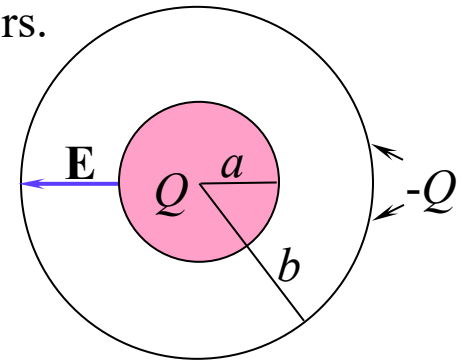
10: Capacitance

• Capacitance in Spherical Capacitor

Consider two concentric spherical conductors, having radii a and b . Equal and opposite charges, Q , are on the inner and outer conductors.

Gauss' Law tells us that electric field will exist only in the region between spheres, and will be given by:

$$\mathbf{E} = E_r \mathbf{a}_r = \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r$$



The potential difference between inner and outer spheres is then:

$$V_0 = - \int_b^a \mathbf{E} \cdot d\mathbf{L} = - \int_b^a \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r \cdot \mathbf{a}_r dr = \frac{Q}{4\pi\epsilon} \left(\frac{1}{a} - \frac{1}{b} \right)$$

and the capacitance is:

$$C = \frac{Q}{V_0} = \frac{4\pi\epsilon}{(1/a) - (1/b)}$$

Note that as $b \rightarrow \infty$ (isolated sphere)

$$C \rightarrow 4\pi\epsilon a$$

Our earth is a huge capacitor!

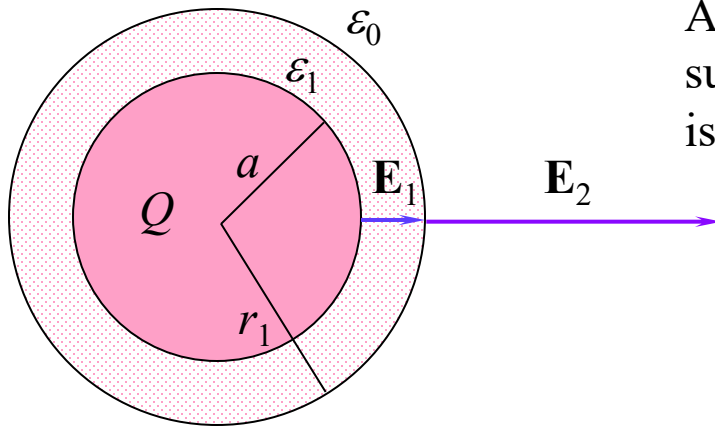


10: Capacitance

• Capacitance in Isolated Sphere with a Dielectric Coating

A conducting sphere of radius a carries charge Q .

A dielectric layer of thickness $r_1 - a$ and of permittivity ϵ_1 surrounds the conductor. Electric field in the two regions is found from Gauss' Law to be:



$$E_r = \frac{Q}{4\pi\epsilon_1 r^2} \quad (a < r < r_1)$$

$$= \frac{Q}{4\pi\epsilon_0 r^2} \quad (r_1 < r)$$

The potential at the sphere surface is (with zero reference at infinity):

$$V_a - V_\infty = - \int_{r_1}^a \frac{Q dr}{4\pi\epsilon_1 r^2} - \int_{\infty}^{r_1} \frac{Q dr}{4\pi\epsilon_0 r^2} = \frac{Q}{4\pi} \left[\frac{1}{\epsilon_1} \left(\frac{1}{a} - \frac{1}{r_1} \right) + \frac{1}{\epsilon_0 r_1} \right] = V_0$$

and the capacitance is:

$$C = \frac{4\pi}{\frac{1}{\epsilon_1} \left(\frac{1}{a} - \frac{1}{r_1} \right) + \frac{1}{\epsilon_0 r_1}}$$

10: Capacitance

- Capacitor with a Two-Layer Dielectric

In this case, we use the fact that $\mathbf{D}$ normal to an interface between two dielectrics will be continuous across the boundary, assuming no surface charge is present there:

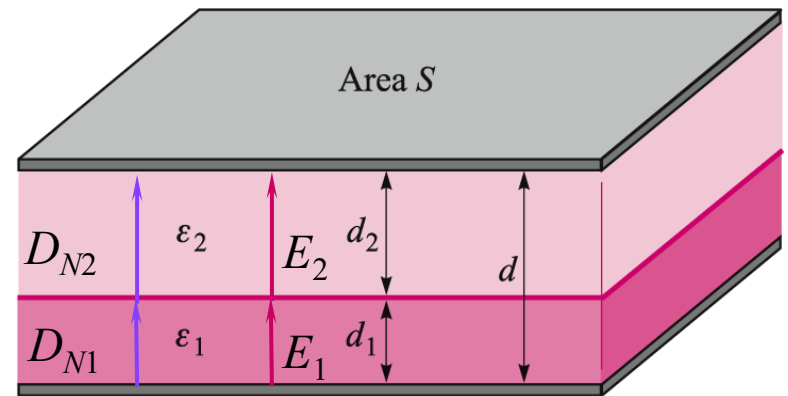
Thus $D_{N1} = D_{N2}$ and therefore: $\epsilon_1 E_1 = \epsilon_2 E_2$

The potential difference between bottom and top plates will be: $V_0 = E_1 d_1 + E_2 d_2$

..which leads to: $E_1 = \frac{V_0}{d_1 + d_2(\epsilon_1/\epsilon_2)}$

The surface charge density on the bottom plate:

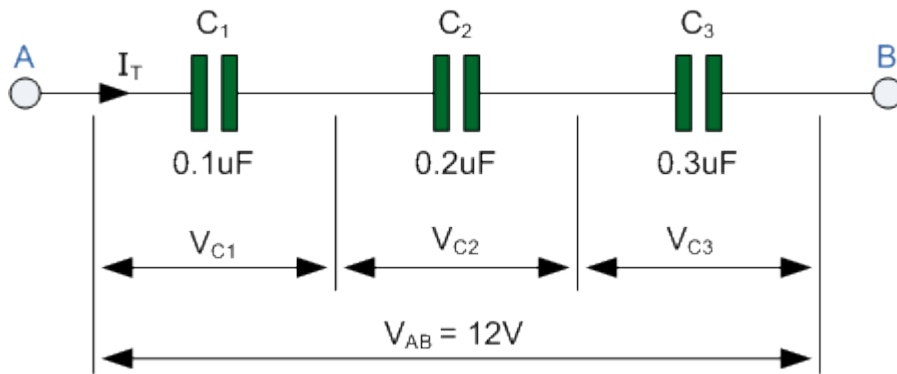
$$\rho_{S1} = D_1 = \epsilon_1 E_1 = \frac{V_0}{\frac{d_1}{\epsilon_1} + \frac{d_2}{\epsilon_2}}$$



The capacitance is then: $C = \frac{Q}{V_0} = \frac{\rho_{S1} S}{V_0} = \frac{1}{\frac{d_1}{\epsilon_1 S} + \frac{d_2}{\epsilon_2 S}} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}}$

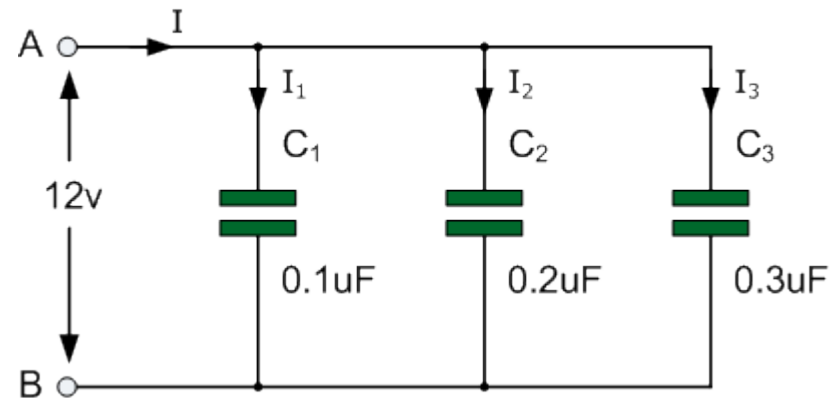
10: Capacitance

- Capacitors in a Series / Parallel Connection

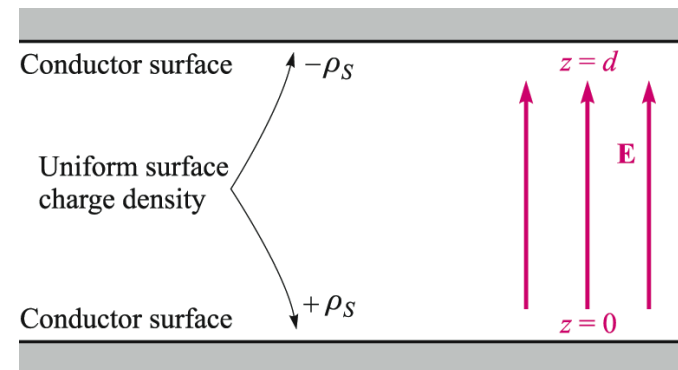


$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$C = \frac{Q}{V_0} = \frac{\epsilon S}{d}$$



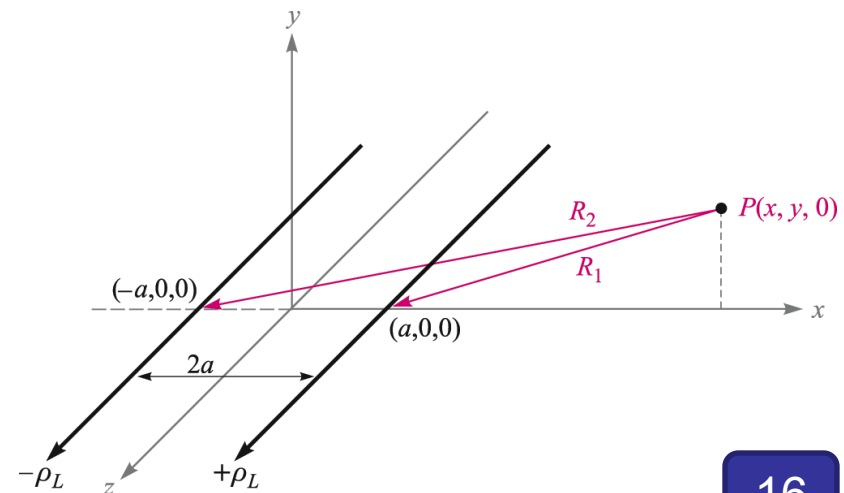
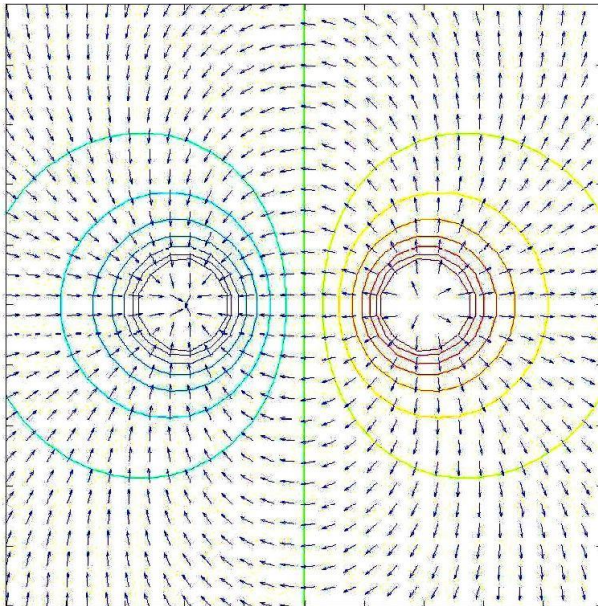
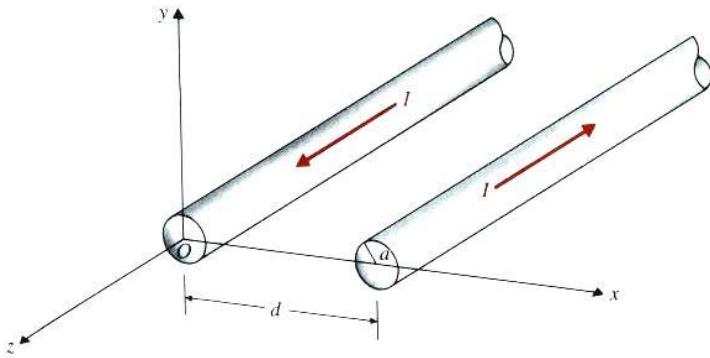
$$C = C_1 + C_2 + C_3$$



10: Capacitance



- Capacitance in Two-Wire Transmission Line



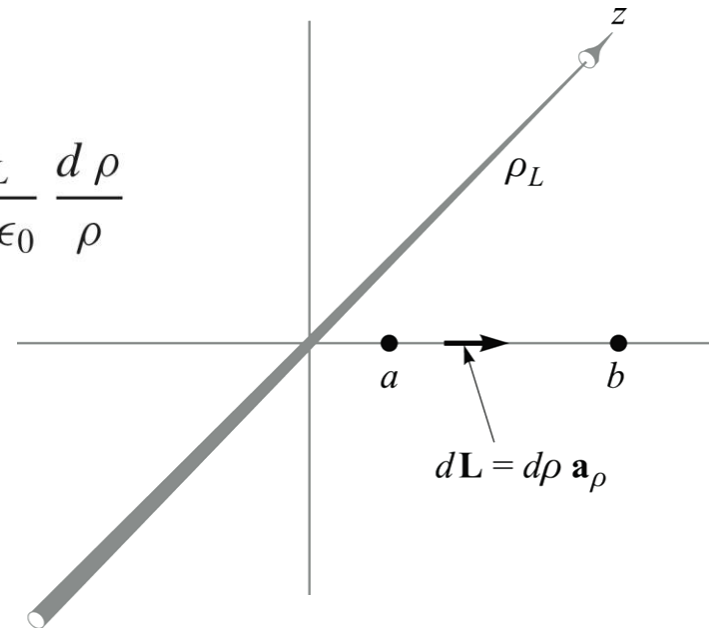
8: Energy and Potential 能量与电位



- Potential difference in a line charge field

$$W = -Q \int_{\text{init}}^{\text{final}} \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho \cdot d\rho \mathbf{a}_\rho = -Q \int_a^b \frac{\rho_L}{2\pi\epsilon_0} \frac{d\rho}{\rho}$$

$$W = -\frac{Q\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$



Thus, the potential difference between points at $\rho = a$ and $\rho = b$ is

$$V_{ab} = \frac{W}{Q} = \frac{\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$

10: Capacitance

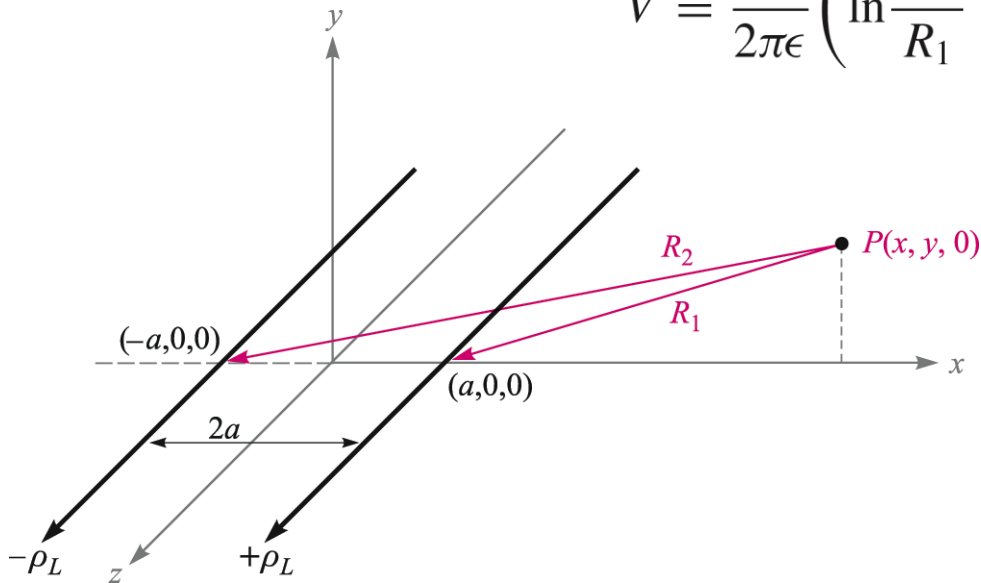
• Capacitance in Two-Wire Transmission Line

We begin with the potential field of a single line charge on the z axis, with a zero reference at $\rho = R_0$

$$V = \frac{\rho_L}{2\pi\epsilon} \ln \frac{R_0}{R}$$

We can use this result to write the potential at point P from two line charges of opposite sign, positioned as shown:

$$V = \frac{\rho_L}{2\pi\epsilon} \left(\ln \frac{R_{10}}{R_1} - \ln \frac{R_{20}}{R_2} \right) = \frac{\rho_L}{2\pi\epsilon} \ln \frac{R_{10} R_2}{R_{20} R_1}$$



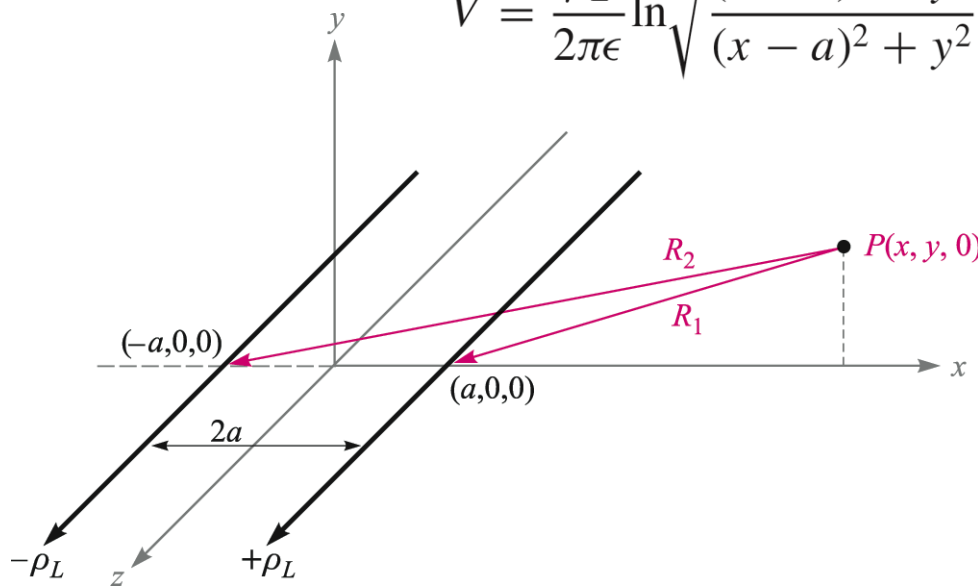
10: Capacitance

- Capacitance in Two-Wire Transmission Line

We now have:
$$V = \frac{\rho_L}{2\pi\epsilon} \left(\ln \frac{R_{10}}{R_1} - \ln \frac{R_{20}}{R_2} \right) = \frac{\rho_L}{2\pi\epsilon} \ln \frac{R_{10} R_2}{R_{20} R_1}$$

We choose $R_{10} = R_{20}$, thus placing the zero reference at equal distances from each line. This surface is the $x = 0$ plane. Expressing R_1 and R_2 in terms of x and y ,

$$V = \frac{\rho_L}{2\pi\epsilon} \ln \sqrt{\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}} = \frac{\rho_L}{4\pi\epsilon} \ln \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}$$



10: Capacitance

- Capacitance in Two-Wire Transmission Line

We now have:
$$V = \frac{\rho_L}{2\pi\epsilon} \ln \sqrt{\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}} = \frac{\rho_L}{4\pi\epsilon} \ln \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}$$

Choose an equipotential surface on which $V = V_1$, and define the dimensionless parameter:

$$K_1 = e^{4\pi\epsilon V_1/\rho_L} \text{ from which we identify: } K_1 = \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}$$

This is the equation of the equipotential surface on which the voltage is V_1

To better identify the surface, expand the squares, and collect terms:

$$x^2 - 2ax \frac{K_1 + 1}{K_1 - 1} + y^2 + a^2 = 0 \quad \Rightarrow \quad \left(x - a \frac{K_1 + 1}{K_1 - 1} \right)^2 + y^2 = \left(\frac{2a\sqrt{K_1}}{K_1 - 1} \right)^2$$

10: Capacitance

- Capacitance in Two-Wire Transmission Line

The equation of an equipotential surface is:

$$\left(x - a \frac{K_1 + 1}{K_1 - 1}\right)^2 + y^2 = \left(\frac{2a\sqrt{K_1}}{K_1 - 1}\right)^2$$

This is the equation of a circle (actually a cylinder), displaced along the x axis by distance h , and having radius b , where

$$h = a \frac{K_1 + 1}{K_1 - 1}$$

and

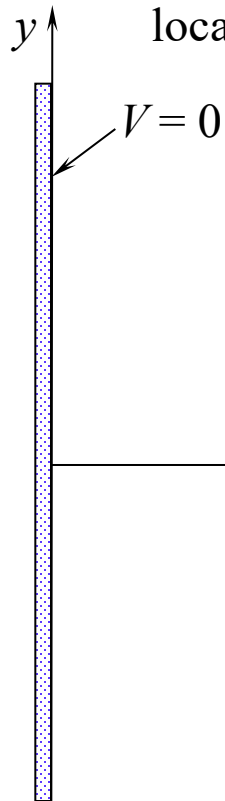
$$b = \frac{2a\sqrt{K_1}}{K_1 - 1}$$

It is all independent of z , as you might expect!

10: Capacitance

• Capacitance in Two-Wire Transmission Line

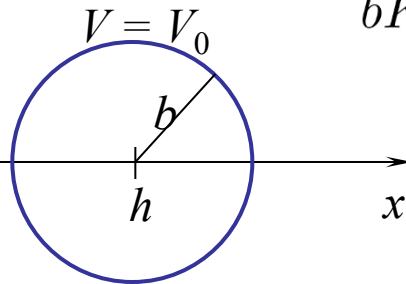
Consider a grounded conductor in the y - z plane, and a conducting cylinder, parallel to the z axis, positioned as shown. The cylinder has radius b and center location h on the x axis, where



$$h = a \frac{K_1 + 1}{K_1 - 1} \quad \text{and} \quad b = \frac{2a\sqrt{K_1}}{K_1 - 1}$$

Eliminate a between the two equations to get the quadratic:

$$bK_1 - 2h\sqrt{K_1} + b = 0$$



whose solution is:

$$\sqrt{K_1} = \frac{h \pm \sqrt{h^2 - b^2}}{b}$$

Choose positive sign because it leads to a positive value for a

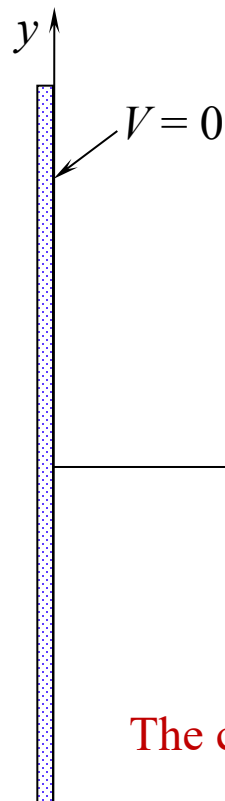
Then, substitute K_1 back into either starting equation to find:

$$a = \sqrt{h^2 - b^2}$$

10: Capacitance

• Capacitance in Two-Wire Transmission Line

The expression $a = \sqrt{h^2 - b^2}$ gives the location of **an equivalent line charge**, ρ_l



From the original definition, we now have $\sqrt{K_1} = e^{2\pi\epsilon V_0/\rho_L}$

$$\text{from which } \rho_L = \frac{4\pi\epsilon V_0}{\ln K_1}$$

Now, given h , b , and V_0 , it is possible to find a , ρ_l , and K_1 .

For a length L in the z direction,

$$C = \frac{\rho_L L}{V_0} = \frac{4\pi\epsilon L}{\ln K_1} = \frac{2\pi\epsilon L}{\ln \sqrt{K_1}}$$

$$\text{So that finally: } C = \frac{2\pi\epsilon L}{\ln[(h + \sqrt{h^2 - b^2})/b]} = \frac{2\pi\epsilon L}{\cosh^{-1}(h/b)}$$

The capacitance between two circular conductors separated by a distance $2h$ is

$$\frac{\pi\epsilon L}{\ln[(h + \sqrt{h^2 - b^2})/b]} = \frac{\pi\epsilon L}{\cosh^{-1}(h/b)}$$

10: Capacitance



- Capacitance in Two-Wire Transmission Line

Numerical Example

Choose $b = 5$ mm, $h = 13$ mm, and $V_0 = 100$ V. Find ρ_l and C .

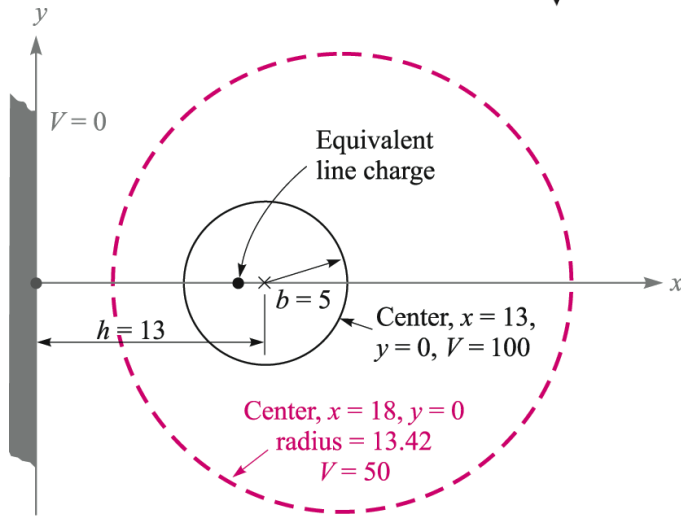
10: Capacitance

• Capacitance in Two-Wire Transmission Line

Numerical Example

Choose $b = 5$ mm, $h = 13$ mm, and $V_0 = 100$ V. Find ρ_L and C .

$$a = \sqrt{h^2 - b^2} = \sqrt{13^2 - 5^2} = 12 \text{ mm}$$



$$\text{Then: } \sqrt{K_1} = \frac{h + \sqrt{h^2 - b^2}}{b} = \frac{13 + 12}{5} = 5$$

Obtain:

$$\rho_L = \frac{4\pi\epsilon V_0}{\ln K_1} = \frac{4\pi \times 8.854 \times 10^{-12} \times 100}{\ln 25} = \underline{3.46 \text{ nC/m}}$$

and

$$C = \frac{2\pi\epsilon}{\cosh^{-1}(h/b)} = \frac{2\pi \times 8.854 \times 10^{-12}}{\cosh^{-1}(13/5)} = \underline{34.6 \text{ pF/m}}$$

$$h = 13, b = 5, \therefore K_1 = 25; \therefore \rho_L = 3.46 \times 10^{-9} \text{ C/m}, \therefore a = 12$$

If $V_1 = 50$, $K_1 = 5$, $h = 18$, $b = 13.42$, ρ_L unchanged

$$C = \frac{2\pi\epsilon_0 L}{\ln 5} = 34.6 \text{ pF/m}$$

10: Capacitance



- Capacitance in Two-Wire Transmission Line

Numerical Example (Cont'd)

In this step, we find the 50-volt equipotential surface

10: Capacitance

• Capacitance in Two-Wire Transmission Line

Numerical Example (Cont'd)

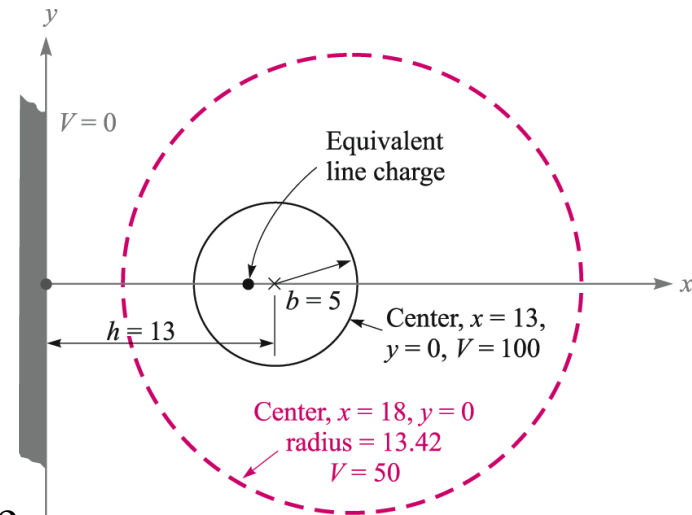
In this step, we find the 50-volt equipotential surface

$$\text{First: } K_1 = e^{4\pi\epsilon V_1/\rho_L} = e^{4\pi \times 8.854 \times 10^{-12} \times 50 / 3.46 \times 10^{-9}} = 5.00$$

$$b = \frac{2a\sqrt{K_1}}{K_1 - 1} = \frac{2 \times 12\sqrt{5}}{5 - 1} = 13.42 \text{ mm}$$

$$h = a \frac{K_1 + 1}{K_1 - 1} = 12 \frac{5 + 1}{5 - 1} = 18 \text{ mm}$$

The resulting surface is the dashed red circle



$$h = 13, b = 5, \therefore K_1 = 25; \therefore \rho_L = 3.46 \times 10^{-9} \text{ C/m}, \therefore a = 12$$

If $V_1 = 50, K_1 = 5, h = 18, b = 13.42, \rho_L$ unchanged

$$C = \frac{2\pi\epsilon_0 L}{\ln 5} = 34.6 \text{ pF/m}$$